

Raj Kumar Nelluri

(201) 554-8009 | rajkumarn2002@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

Python • SQL • MySQL • AWS SageMaker • LangChain • LLMs • Tableau • FastAPI • Machine Learning • A/B Testing • Git

PROFESSIONAL SUMMARY

Machine Learning Engineer with hands-on experience building, evaluating, and **deploying ML models using Python, scikit-learn, XGBoost, TensorFlow, and AWS**. Built end-to-end ML workflows for customer churn prediction, fraud detection, and demand forecasting using structured, behavioral, and text-derived data. Experienced in feature engineering, model tuning, pipeline automation, REST API deployment, and monitoring. Strong foundation in machine learning workflows, Python programming, and cloud-based ML systems, with a focus on shipping practical models that improve business decision-making.

EDUCATION

Pace University

Master of Science in Computer Science

Amrita Vishwa Vidyapeetham

Bachelor of Technology in Artificial Intelligence

New York, NY

Sep 2023 – April 2025

Bengaluru, India

July 2019 – April 2023

PROFESSIONAL EXPERIENCE

Enterprise RAG Chatbot – LLM-Powered Document Intelligence | *Python, LangChain, OpenAI,* 2026

ChromaDB, Streamlit, Docker, AWS EC2 | [GitHub](#) | [Live Demo](#)

- Engineered production-grade **Retrieval-Augmented Generation (RAG)** pipeline using **LangChain** and OpenAI API – ingests PDF documents, chunks via RecursiveCharacterTextSplitter (1000-char / 200-overlap), and persists embeddings (text-embedding-3-small, 1536-dim) in a ChromaDB vector store
- Implemented **MMR (Maximum Marginal Relevance)** retrieval to fetch 20 candidate chunks and re-rank to top-5, balancing relevance and diversity to derive **actionable insights from unstructured documents** using **LLMs**
- Designed metadata-injected context pipeline preserving source filename and page number in every LLM prompt, enforcing mandatory [**SOURCE N: filename — Page**] citations on all responses and eliminating hallucination on out-of-scope queries; supports both pre-loaded document library and upload-any-PDF mode
- Dockerized** the application and deployed to **AWS EC2** (t3.micro, Ubuntu 24.04) with a persistent ChromaDB named volume and **--restart unless-stopped**, delivering a live public demo running 24/7

Customer Churn Prediction & Engagement Strategy | *Python, XGBoost, SageMaker, Lambda,* 2025

Comprehend, CloudWatch | [GitHub](#)

- Designed end-to-end **ML lifecycle** workflow on AWS – S3 ingestion → Lambda ETL orchestration → SageMaker managed training – automating the full pipeline from problem definition through **deployed production model**
- Extracted NLP features from 7,000+ unstructured customer support tickets using AWS Comprehend, transforming raw text into structured sentiment scores and key-phrase signals to enrich **customer engagement modeling**
- Deployed **XGBoost classifier as a SageMaker real-time inference endpoint**; tuned hyperparameters and optimized decision threshold to achieve 0.67 recall, enabling proactive **customer outreach** for high-risk churn segments
- Implemented **automated model monitoring** using CloudWatch alarms and EventBridge rules, triggering retraining pipelines upon data drift detection – ensuring 24/7 operational reliability of the production scoring system

Insurance Fraud Detection – Automated ML Pipeline | *Python, Scikit-learn, Lambda, Kinesis, Capstone – Pace* | 2024

RDS, SageMaker

- Architected 4-stage cloud-native **automated ML pipeline** – S3 data lake ingestion → Lambda ETL trigger → Kinesis distributed stream processing → SageMaker inference – eliminating manual intervention in end-to-end claim scoring
- Engineered ETL transformation layer in **Python** to validate schema, enforce data quality checks, impute missing values, and normalize 15,000+ insurance claim records across 30+ features before model ingestion
- Trained Random Forest and Gradient Boosting classifiers with 5-fold cross-validation and hyperparameter tuning; optimized classification threshold achieving approximately 90% accuracy on held-out test data
- Persisted fraud scores to **AWS RDS** data warehouse, enabling analyst teams to query ML predictions via **SQL** without re-executing the training pipeline – directly reducing manual analytical workload

Retail Sales Forecasting & Demand Analytics | *Python, XGBoost, Pandas, NumPy, AWS S3, Sage-* 2025

Maker | [GitHub](#)

- Built **scalable batch ETL pipeline** using Python and AWS S3 to ingest, aggregate, and transform 500,000+ retail transactions into structured daily time-series datasets, enabling **data-driven demand forecasting** for business planning
- Engineered 10+ temporal features – lag variables, 7/30-day rolling averages, and holiday indicators – capturing weekly seasonality and **business KPI trends** across a 2-year sales history
- Trained and deployed XGBoost regression model via **SageMaker real-time inference endpoint** with hyperparameter tuning; evaluated using MAE and RMSE on held-out validation set to ensure forecast accuracy

Crypto Price Forecasting API | *Python, XGBoost, TensorFlow, LSTM, CNN, Flask* *Capstone – Amrita* | 2023

- Benchmarked 3 ML architectures (LSTM, CNN, XGBoost) on chronological train-test split; selected XGBoost based on lowest MAE/RMSE, demonstrating **hypothesis-driven model selection** methodology
- Deployed production model as a **Flask REST API** serving live 7-day price forecasts via JSON endpoint powering a web dashboard with sub-second response latency

3D Face Generation (Neural Radiance Fields) | *Python, JAX, TensorFlow, OpenCV, COLMAP* *Capstone – Amrita* | 2023

- Implemented Deformable NeRF in JAX/TensorFlow for 3D face reconstruction from monocular video; optimized GPU memory consumption by over 30% through iterative batch tuning, enabling training on standard hardware

CERTIFICATIONS

AWS Certified Cloud Practitioner – Amazon Web Services	2025
Adobe Journey Optimizer Foundation – Adobe	2025
Complete Python Developer – Udemy	2023

TECHNICAL SKILLS

Programming: Python, SQL, MySQL
AI & LLMs: LangChain, OpenAI API, RAG Pipelines, Prompt Engineering, Vector Databases, ChromaDB, Embeddings, Semantic Search
Machine Learning: Scikit-learn, XGBoost, TensorFlow, Keras, Feature Engineering, Hyperparameter Tuning, A/B Testing, Statistical Analysis, ML Lifecycle
Data Engineering: ETL/ELT Pipelines, Batch Processing, Distributed Stream Processing, Data Validation, Data Quality, Schema Enforcement
Cloud (AWS): S3, SageMaker, Lambda, Kinesis, RDS, Glue, EC2, CloudWatch, EventBridge
APIs & Deployment: REST APIs, Flask, FastAPI, Streamlit, Docker, Real-Time Inference, Batch Inference, Model Monitoring
Visualization & Analytics: Tableau, Power BI, Matplotlib
Tools: Git, GitHub, Jupyter Notebook, VS Code